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Introduction / Background:

- The COVID-19 pandemic had a profound impact on consumer behaviour and household spending patterns across the United Kingdom. Many families altered their financial priorities, reducing spending on non-essential goods such as recreation, transport, and clothing, while increasing expenditure on essentials like food, housing, and utilities.
- This project aims to analyze how **family spending behavior changed before and after the COVID-19 pandemic**, focusing on trends between **2019 and 2022**. The analysis will use publicly available government data from the **Office for National Statistics (ONS)** to uncover key differences in expenditure categories, helping to understand which sectors were most affected and how consumer confidence evolved during and after the pandemic.
- I selected this topic because it reflects a significant social and economic shift that impacted every household. Understanding these spending trends can help policymakers, businesses, and researchers anticipate future consumer behavior in times of economic uncertainty.

Research Question:

- How did household spending patterns in the United Kingdom change before and after the COVID-19 pandemic (2019–2022)?

Model Selected:

- I have selected **Time Series Analysis** as the analytical model for this project. This model is appropriate because it helps identify **trends, seasonality, and changes over time** in household spending categories such as food, housing, recreation, and transport. By comparing expenditure trends from **2019 (pre-COVID)** to **2022 (post-COVID)**, the model can illustrate how consumption behaviour evolved through different phases of the pandemic and recovery period.
- Visualisations such as **line charts, trend plots, and forecast projections** will be used in **Power BI** to effectively communicate these temporal changes.

Dataset:

- The dataset used for this project is obtained from the **Office for National Statistics (ONS)** — a reliable and government-approved source, <https://www.ons.gov.uk/peoplepopulationandcommunity/personalandhouseholdfinances/expenditure/datasets/familyspendingworkbook1detailedexpenditureandtrends> <https://www.ons.gov.uk/peoplepopulationandcommunity/personalandhouseholdfinances/expenditure/datasets/livingcostsandfoodsurveytechnicalreportdatables>
- The dataset includes household expenditure by category and year, measured in average weekly spending per household. It covers key variables such as: Food and non-alcoholic drinks, Housing, fuel and power, Transport, Recreation and culture, Restaurants and hotels

Family Spending

Workbook 1 - Detailed Expenditure and Trends

Contents:

- Annex Tables in reports on the Living Costs and Food Survey, financial year ending 2001 to financial year ending 2020
- Table A1 - Detailed expenditure with full-method standard errors
- Table A2 - Expenditure on food by place of purchase
- Table A3 - Expenditure on clothing and footwear by place of purchase
- Table A6 - Detailed expenditure by gross income decile
- Table A11 - Detailed expenditure by age of Household Reference Person
- Table 3.1 - Detailed household expenditure by disposable income decile group
- Table 3.1E Detailed household expenditure by disposable equivalised income decile group (OECD-modified scale)
- Table 3.2 - Percentage of total expenditure by disposable income decile group
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- Table 4.1 - COICOP main items, 2001/02 to 2019/20
- Table 4.2 - COICOP as a percentage of total expenditure, 2001/02 to 2019/20
- Table 4.3 - Household expenditure, 2001/02 to 2019/20 COICOP based on current prices
- Background notes

ONS Social Surveys
Office for National Statistics

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Percentage of all households

Table 14 - Percentage standard errors and confidence intervals of expenditure

UK, financial year ending 2024

Commodity or service	Weighted average weekly household expenditure (£)	Full method standard error	Households recording expenditure	Percentage of all households	95% confidence interval	
					Lower limit (£)	Upper limit (£)
Food & non-alcoholic drinks	70.50	0.8	4,180	99	69.40	71.60
Alcoholic drink, tobacco & narcotics	11.10	2.8	2,320	55	10.50	11.70
Clothing & footwear	17.50	4.1	2,190	52	16.10	18.90
Housing(fuel), fuel & power	113.30	1.2	4,200	100	110.60	116.00
Household goods & services	38.70	2.8	3,830	91	36.60	40.80
Health	7.60	6.7	2,330	55	6.60	8.60
Transport	88.20	2.2	3,710	88	84.40	92.00
Communication	22.20	1.2	4,130	98	21.70	22.70
Recreation & culture	72.80	2.4	4,120	98	69.40	76.20
Education	5.10	24.8	200	5	2.60	7.60
Restaurants & hotels	44.40	1.8	3,270	78	42.80	46.00
Miscellaneous goods and services	45.60	1.8	4,110	98	44.00	47.20
All expenditure groups	537.00	1.0	4,210	100	526.50	547.50
Other expenditure items	86.30	7.2	4,060	96	81.70	90.90
Total expenditure	623.30	1.1	4,210	100	609.90	636.70

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10°C Mobility summary

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Table A2

Expenditure on food and non-alcoholic drinks by place of purchase

UK, financial year ending 2020

Commodity or service	Large supermarket chains ¹			Other outlets			Internet expenditure ²		
	Average weekly expenditure per household (£)	Households recording expenditure (thousands)	Percentage of all households	Average weekly expenditure per household (£)	Households recording expenditure (thousands)	Percentage of all households	Average weekly expenditure per household (£)	Households recording expenditure (thousands)	Percentage of all households
Food and non-alcoholic drinks	48.70	5,353	5,380	10.40	289	4,680	4.70	130	610
1.1 Bread, rice and cereals	44.00	1,240	5,200	9.40	261	4,590	4.20	117	590
1.1.1 Bread, rice and cereals	4.00	120	5,070	0.80	22	1,500	0.40	11	440
1.1.2 Pasta products	0.40	10	2,070	0.10	3	270	0.00	1	210
1.1.3 Other food products	3.60	80	4,820	0.70	18	1,730	0.00	6	340
1.1.4 Pasta (excludes)	0.30	24	2,110	0.00	1	100	0.10	2	150
1.1.5 Beef, lamb, chicken or turkey	1.20	27	2,040	0.10	12	430	0.10	7	170
1.1.6 Pork, chicken, chicken or turkey	0.40	11	880	0.10	3	190	0.00	1	60
1.1.7 Lamb, chicken, chicken or turkey	0.40	10	480	0.10	9	160	0.00	1	40
1.1.8 Other meat, chicken or turkey	1.80	60	2,610	0.60	13	610	0.00	6	390
1.1.9 Bacon and ham	0.70	19	2,000	0.10	3	310	0.10	1	140
1.1.10 Other meat and meat preparations	5.20	140	4,400	0.90	20	1,410	0.40	12	290
1.1.11 Fish and fish products	2.00	69	2,200	0.50	12	490	0.20	6	270
1.1.12 Eggs	1.60	44	1,360	0.40	10	1,300	0.10	3	240
1.1.13 Other meat and fish	1.80	51	3,730	0.20	5	610	0.00	5	330
1.1.14 Eggs	0.60	17	2,750	0.10	3	630	0.10	2	240
1.1.15 Other meat products	2.20	59	4,000	0.10	4	540	0.20	6	240
1.1.16 Butter	0.40	11	1,270	0.00	1	170	0.00	1	110
1.1.17 Other products, other vegetable fats and oils	0.40	11	1,260	0.00	1	230	0.00	1	170
1.1.18 Condensed milk and milk	0.30	8	1,070	0.10	3	130	0.00	1	80
1.1.19 Cream and milk	3.40	93	4,430	0.40	10	830	0.30	8	370
1.1.20 Other food, chicken or turkey	0.40	11	1,000	0.10	1	170	0.00	1	170

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10°C Mobility summary

References:

- [1] O. F. N. Statistics, "Family Spending in the UK: April 2019 to March 2020," Office For National Statistics, 2020.
- [2] G. E. P. Box, G. M. Jenkins and G. C. Reinsel, Time Series Analysis: Forecasting and Control, Hoboken, NJ: Wiley, 2008.